# Coordinate Geometry

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Distance and midpoint formulas**

Find the distance between $(2, -1)$ and $(10, 5)$.

Distance $= \sqrt{(10-2)^2 + (5-(-1))^2} = \sqrt{64+36} = \sqrt{100} = 10$.

**2** **[Foundational]** • **Target: Distance and midpoint formulas**

Find the midpoint of the segment connecting $(-4, 7)$ and $(6, -3)$.

Midpoint $= \left(\frac{-4+6}{2}, \frac{7+(-3)}{2}\right) = (1, 2)$.

**3** **[Foundational]** • **Target: Slope and equations of lines**

Find the slope of the line through $(1, 4)$ and $(5, -8)$.

Slope $= \frac{-8-4}{5-1} = \frac{-12}{4} = -3$.

**4** **[Foundational]** • **Target: Slope and equations of lines**

Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$, in slope-intercept form.

$y-(-3) = \frac{1}{2}(x-2)$, so $y+3 = \frac{1}{2}x - 1$, giving $y = \frac{1}{2}x - 4$.

**5** **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 6)$.

Parallel lines share the same slope: $m = 4$. Since $(0, 6)$ is the $y$-intercept, the equation is $y = 4x + 6$.

**6** **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.

The perpendicular slope is the negative reciprocal: $m = \frac{3}{2}$. $y - 1 = \frac{3}{2}(x-4)$, so $y = \frac{3}{2}x - 6 + 1 = \frac{3}{2}x - 5$.

**7** **[Foundational]** • **Target: The equation of a circle**

Write the equation of a circle with center $(3, -2)$ and radius 7.

$(x-3)^2 + (y+2)^2 = 49$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Reflections, translations, and symmetry

Find the coordinates of the point $(5, -3)$ after a reflection over the $x$-axis, then after a reflection over the $y$-axis.

Reflection over the $x$-axis: $(5, 3)$ (the $y$-coordinate's sign flips). Reflection over the $y$-axis (applied to the original point): $(-5, -3)$ (the $x$-coordinate's sign flips).

**9** [Intermediate] • Target: Finding area using the coordinate plane

Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$.

Since the triangle has a right angle at the origin, with legs along the axes: Area $= \frac{1}{2}(6)(4) = 12$.

**10** [Intermediate] • Target: Finding area using the coordinate plane

A rectangle has vertices $(1, 1)$, $(1, 5)$, $(7, 5)$, and $(7, 1)$. Find its area and perimeter.

Width: $7 - 1 = 6$. Height: $5 - 1 = 4$. Area $= 6 \times 4 = 24$. Perimeter $= 2(6 + 4) = 20$.

**11** [Intermediate] • Target: Determining if points are collinear

Determine whether the points $(1, 2)$, $(3, 6)$, and $(5, 10)$ are collinear, by comparing slopes.

Slope from $(1, 2)$ to $(3, 6)$: $\frac{6-2}{3-1} = \frac{4}{2} = 2$. Slope from $(3, 6)$ to $(5, 10)$: $\frac{10-6}{5-3} = \frac{4}{2} = 2$. Since both slopes are equal, the three points are collinear.

**12** [Intermediate] • Target: Finding the intersection of two lines

Find the point of intersection of $y = 2x - 3$ and $y = -x + 9$.

Set equal: $2x - 3 = -x + 9$, so $3x = 12$, giving $x = 4$. Then $y = 2(4) - 3 = 5$. Intersection: $(4, 5)$.

**13** [Intermediate] • Target: Combining coordinate geometry concepts

This question has two parts. Triangle $ABC$ has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 6)$. (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.

(a) $AB = \sqrt{(4-0)^2 + (0-0)^2} = 4$. $AC = \sqrt{(2-0)^2 + (6-0)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$. $BC = \sqrt{(2-4)^2 + (6-0)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$. (b) Since $AC = BC = 2\sqrt{10}$ (two sides are equal in length), the triangle is isosceles.

**14** [Intermediate] • Target: Graphing inequalities in the coordinate plane

Determine whether the point $(3, 1)$ satisfies the inequality $y < 2x - 3$.

Substitute: $1 < 2(3) - 3 = 3$? True. The point satisfies the inequality.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: Graphing inequalities in the coordinate plane • Cross-Lesson: 5.01 Plane Geometry

A region is defined by $y \geq x + 1$. Determine whether $(0, 0)$ lies in this region.

Substitute: $0 \geq 0 + 1 = 1$? False. The point $(0, 0)$ does not lie in the region.

**16** [Challenging] • Target: The equation of a parabola in the coordinate plane • Cross-Lesson: 6.01 Trigonometry

Write the equation of a parabola with vertex $(2, -3)$ that passes through $(4, 5)$.

Vertex form: $y = a(x - 2)^2 - 3$. Substitute $(4, 5)$: $5 = a(4 - 2)^2 - 3$, so $5 = 4a - 3$, giving $a = 2$. Equation: $y = 2(x - 2)^2 - 3$.

**17** [Challenging] • Target: Finding points that divide a segment in a given ratio • Cross-Lesson: 7.01 Statistics & Probability

Find the point that divides the segment from $(1, 2)$ to $(9, 10)$ in a ratio of $1 : 3$ (closer to the first point).

Using the section formula for a $1 : 3$ ratio: $x = 1 + \frac{1}{4}(9 - 1) = 1 + 2 = 3$. $y = 2 + \frac{1}{4}(10 - 2) = 2 + 2 = 4$. Point: $(3, 4)$.

**18** [Challenging] • Target: A multi-part problem combining lines and circles • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

This question has two parts. A circle has equation $(x - 2)^2 + (y + 1)^2 = 25$. (a) State the center and radius. (b) Determine whether the point $(5, 3)$ lies inside, on, or outside the circle, by computing its distance from the center.

(a) Center: $(2, -1)$. Radius: $\sqrt{25} = 5$. (b) Distance from center to $(5, 3)$: $\sqrt{(5 - 2)^2 + (3 - (-1))^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Since this distance equals the radius, the point $(5, 3)$ lies exactly ON the circle.

**19** [Challenging] • Target: Systems of a line and a circle • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the points of intersection of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$.

Substitute: $x^2 + (x + 1)^2 = 25$, so $x^2 + x^2 + 2x + 1 = 25$, giving $2x^2 + 2x - 24 = 0$, or $x^2 + x - 12 = 0$. Factor: $(x + 4)(x - 3) = 0$, so $x = -4$ or $x = 3$. Corresponding $y$-values: for $x = -4$, $y = -3$; for $x = 3$, $y = 4$. Intersection points: $(-4, -3)$ and $(3, 4)$.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.

The perpendicular slope is the negative reciprocal: $m = \frac{3}{2}$. $y - 1 = \frac{3}{2}(x - 4)$, so $y = \frac{3}{2}x - 6 + 1 = \frac{3}{2}x - 5$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Distance and midpoint formulas

Find the distance between $(8, 8)$ and $(18, 32)$.

**a.** 27

**b.** 34

**c.** 25

**d.** 26 ✓

Distance $= \sqrt{(18-(8))^2 + (32-(8))^2} = \sqrt{(10)^2 + (24)^2} = \sqrt{676} = 26.$

**22** [Foundational] • Target: Slope and equations of lines

Find the distance between $(-2, 7)$ and $(-29, -29)$.

**a.** 63

**b.** 46

**c.** 45 ✓

**d.** 44

Distance $= \sqrt{(-29-(-2))^2 + (-29-(7))^2} = \sqrt{(-27)^2 + (-36)^2} = \sqrt{2025} = 45.$

**23** [Foundational] • Target: Parallel and perpendicular lines

Find the distance between $(0, 2)$ and $(8, 17)$.

**a.** 16

**b.** 18

**c.** 17 ✓

**d.** 23

Distance $= \sqrt{(8-(0))^2 + (17-(2))^2} = \sqrt{(8)^2 + (15)^2} = \sqrt{289} = 17.$

**24** [Foundational] • Target: The equation of a circle

Find the distance between $(-2, 4)$ and $(5, -20)$.

**a.** 26

**b.** 24

**c.** 31

**d.** 25 ✓

Distance $= \sqrt{(5-(-2))^2 + (-20-(4))^2} = \sqrt{(7)^2 + (-24)^2} = \sqrt{625} = 25.$

**25** [Foundational] • Target: Reflections, translations, and symmetry

Find the distance between $(6, -3)$ and $(-12, -27)$.

**a.** 29

**b.** 42

**c.** 30 ✓

**d.** 31

Distance $= \sqrt{(-12-(6))^2 + (-27-(-3))^2} = \sqrt{(-18)^2 + (-24)^2} = \sqrt{900} = 30.$

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**26** [Foundational] • Target: Finding area using the coordinate plane

Find the distance between $(-7, -6)$ and $(17, -51)$.

**a.** 52

**b.** 50

**c.** 51 ✓

**d.** 69

Distance $= \sqrt{(17-(-7))^2 + (-51-(-6))^2} = \sqrt{(24)^2 + (-45)^2} = \sqrt{2601} = 51$.

**27** [Foundational] • Target: Determining if points are collinear

Find the distance between $(3, -5)$ and $(-17, 43)$.

**a.** 52 ✓

**b.** 53

**c.** 68

**d.** 51

Distance $= \sqrt{(-17-(3))^2 + (43-(-5))^2} = \sqrt{(-20)^2 + (48)^2} = \sqrt{2704} = 52$.

**28** [Foundational] • Target: Finding the intersection of two lines

Find the distance between $(8, -5)$ and $(-19, -125)$.

**a.** 122

**b.** 123 ✓

**c.** 124

**d.** 147

Distance $= \sqrt{(-19-(8))^2 + (-125-(-5))^2} = \sqrt{(-27)^2 + (-120)^2} = \sqrt{15129} = 123$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Combining coordinate geometry concepts

Find the midpoint of the segment connecting $(9, -5)$ and $(1, 11)$.

**a.** $(5, 4)$

**b.** $(6, 3)$

**c.** $(5, 3)$ ✓

**d.** $(-5, 3)$

Midpoint $= \left(\frac{9+1}{2}, \frac{-5+11}{2}\right) = (5, 3)$.

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**30** [Intermediate] • Target: Graphing inequalities in the coordinate plane

Find the midpoint of the segment connecting $(-6, 0)$ and $(4, 12)$.

**a.** $(1, 6)$

**b.** $(0, 6)$

**c.** $(-1, 6)$ ✓

**d.** $(-1, 7)$

Midpoint $= \left(\frac{-6+4}{2}, \frac{0+12}{2}\right) = (-1, 6)$.

**31** [Intermediate] • Target: The equation of a parabola in the coordinate plane

Find the midpoint of the segment connecting $(2, 8)$ and $(10, -4)$.

**a.** $(6, 3)$

**b.** $(6, 2)$ ✓

**c.** $(7, 2)$

**d.** $(-6, 2)$

Midpoint $= \left(\frac{2+10}{2}, \frac{8-4}{2}\right) = (6, 2)$.

**32** [Intermediate] • Target: Finding points that divide a segment in a given ratio

Find the midpoint of the segment connecting $(7, 3)$ and $(-11, -11)$.

**a.** $(-2, -3)$

**b.** $(-1, -4)$

**c.** $(2, -4)$

**d.** $(-2, -4)$ ✓

Midpoint $= \left(\frac{7-11}{2}, \frac{3-11}{2}\right) = (-2, -4)$.

**33** [Intermediate] • Target: A multi-part problem combining lines and circles

Find the midpoint of the segment connecting $(8, 1)$ and $(0, -7)$.

**a.** $(5, -3)$

**b.** $(4, -3)$ ✓

**c.** $(4, -2)$

**d.** $(-4, -3)$

Midpoint $= \left(\frac{8}{2}, \frac{1-7}{2}\right) = (4, -3)$.

**34** [Intermediate] • Target: Systems of a line and a circle

Find the midpoint of the segment connecting $(-7, -3)$ and $(3, 1)$.

**a.** $(-2, 0)$

**b.** $(-2, -1)$ ✓

**c.** $(2, -1)$

**d.** $(-1, -1)$

Midpoint $= \left(\frac{-7+3}{2}, \frac{-3+1}{2}\right) = (-2, -1)$.

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**35** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the midpoint of the segment connecting $(12, 9)$ and $(-4, 11)$.

**a.** $(-4, 10)$

**b.** $(4, 11)$

**c.** $(5, 10)$

**d.** $(4, 10)$ ✓

Midpoint $= \left(\frac{12-4}{2}, \frac{9+11}{2}\right) = (4, 10)$.

**36** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the radius of the circle $x^2 + y^2 + 16x - 18y + 81 = 0$, by completing the square.

**a.** 8 ✓

**b.** 7

**c.** 64

**d.** 9

Completing the square: $(x + 8)^2 + (y - 9)^2 = 64$. Center: $(-8, 9)$. Radius: $\sqrt{64} = 8$.

**37** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the radius of the circle $x^2 + y^2 - 16y + 15 = 0$, by completing the square.

**a.** 49

**b.** 7 ✓

**c.** 8

**d.** 6

Completing the square: $(x)^2 + (y - 8)^2 = 49$. Center: $(0, 8)$. Radius: $\sqrt{49} = 7$.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Find the radius of the circle $x^2 + y^2 - 8x - 16y - 41 = 0$, by completing the square.

**a.** 11 ✓

**b.** 121

**c.** 12

**d.** 10

Completing the square: $(x - 4)^2 + (y - 8)^2 = 121$. Center: $(4, 8)$. Radius: $\sqrt{121} = 11$.

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**39** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the equation of the line perpendicular to $y = -\frac{1}{4}x$ passing through $(0, 5)$, in slope-intercept form.

**a.** $y = -\frac{1}{4}x + 5$

**b.** $y = 4x + 6$

**c.** $y = -4x + 5$

**d.** $y = 4x + 5$ ✓

The perpendicular slope is the negative reciprocal: $m = 4$. $y - (5) = 4(x - 0)$, giving $y = 4x + 5$.

**40** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the equation of the line perpendicular to $y = -\frac{2}{3}x$ passing through $(3, -1)$, in slope-intercept form.

**a.** $y = \frac{3}{2}x - \frac{9}{2}$

**b.** $y = -\frac{2}{3}x - \frac{11}{2}$

**c.** $y = -\frac{3}{2}x - \frac{11}{2}$

**d.** $y = \frac{3}{2}x - \frac{11}{2}$ ✓

The perpendicular slope is the negative reciprocal: $m = \frac{3}{2}$. $y - (-1) = \frac{3}{2}(x - 3)$, giving $y = \frac{3}{2}x - \frac{11}{2}$.

**41** [Challenging] • Target: Distance and midpoint formulas • Cross-Lesson: 3.01 Functions & Graphs

Find the point of intersection of $y = 3x - 17$ and $y = -2x - 2$.

**a.** $(3, -8)$ ✓

**b.** $(4, -8)$

**c.** $(-3, -8)$

**d.** $(3, 8)$

Set equal: $3x - 17 = -2x - 2$, giving $x = 3$. Then $y = 3(3) - 17 = -8$. Intersection: $(3, -8)$.

**42** [Challenging] • Target: Slope and equations of lines • Cross-Lesson: 5.01 Plane Geometry

Find the point of intersection of $y = x - 12$ and $y = 4x - 39$.

**a.** $(9, 3)$

**b.** $(-9, -3)$

**c.** $(9, -3)$ ✓

**d.** $(10, -3)$

Set equal: $x - 12 = 4x - 39$, giving $x = 9$. Then $y = (9) - 12 = -3$. Intersection: $(9, -3)$.

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**43** [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 6.01 Trigonometry

Find the point of intersection of $y = 5x + 38$ and $y = -3x - 10$.

**a.** $(-5, 8)$

**b.** $(6, 8)$

**c.** $(-6, -8)$

**d.** $(-6, 8)$ ✓

Set equal: $5x + 38 = -3x - 10$, giving $x = -6$. Then $y = 5(-6) + 38 = 8$. Intersection: $(-6, 8)$.

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Which of the following points lies on the line that passes through $(1, 4)$ and has a slope of $-3$?

**a.** $(3, -2)$ ✓

**b.** $(2, 2)$

**c.** $(4, -4)$

**d.** $(0, 6)$

Line equation: $y - 4 = -3(x - 1) \Rightarrow y = -3x + 7$. Testing $(3, -2)$: $-3(3) + 7 = -9 + 7 = -2$ (true).

**45** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A circle has the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of this circle?

**a.** 4

**b.** 5 ✓

**c.** 6

**d.** 10

Complete the square: $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 9 + 16 \Rightarrow (x - 3)^2 + (y + 4)^2 = 25$. Radius $r = \sqrt{25} = 5$.

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